

Muhammad Mahdi Qasimi

Doctoral Researcher, Faculty Engineering and
Natural Sciences, Tampere University

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RESEARCH FIELDS

Cold spray coatings, coating characterization, confined space deposition, particle impact behavior and process optimization.

EMPLOYMENT

- 2026 – Tampere University, Materials Characterization/Coating Technologies Group –
Doctoral Researcher
- 2025 Tampere University, Coating Technologies Group – Research assistant

EDUCATION

- 2026 – Tampere University
Doctor of Science in Materials Science and Engineering
- 2023 – 2025 Tampere University
Master of Science in Materials Science and Engineering, with distinction
- 2018 – 2022 University of the Punjab
Bachelor of Engineering in Metallurgy and Materials Engineering

RESEARCH OUTPUT

Investigation of In-flight Behavior and Coating Property Correlation in Internal Diameter Cold Spray – Manuscript (not published)

In-flight Behavior and Coating Property Correlation for IDSC, [link](#)
International Cold Spray Conference & Expo 2026, Chennai, India – Conference Talk

Development of data-driven predictive tool for coating quality assessment, as an experimental tool for analyzing particle in-flight behavior spray and its correlation with coating properties in cold spray – Developed Tool

Estimation of Deformation in Dome-shape Structure Made of FGM for Aerospace Applications
All Pakistan Materials Engineering Competition 2022, UET Lahore, Pakistan – Poster

EDUCATION

- 2022 **Winner**
Comp 1.0, All Pakistan Materials Engineering Competition (simulation)
- 2022 **Best Poster Award**
Final year research project, University of the Punjab,
- 2021 **Competition Award**
NACE Pakistan

TECHNICAL SKILLS

Cold Spray & Diagnostics — Cold spray system operation; particle velocity measurement; camera calibration

Materials Characterization & Testing — SEM; Optical microscopy; hardness testing; metallographic sample preparation; corrosion and electrochemical testing (EIS)

Computational — Python (data analysis, scikit-learn); ANSYS (finite element modelling); SolidWorks; OriginLab

Research Methods — Experimental design; data analysis and interpretation; scientific writing and reporting

MOOCs & TRAINING

MOOCs — Machine Learning Specialization, Coursera (in progress); Introduction to High-Throughput Materials Development, Coursera; Materials Data Science and Informatics, Coursera

NDT Training — Introductory training in VT, MT, PT and UT, supervised by Level III NDT personnel

SUPERVISORS

Prof. Minnamari Vippola — minnamari.vippola@tuni.fi

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